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ORIGINAL RESEARCH

A Study of the Dissemination of Malware and the Enhancement of the Lifespan of Rechargeable Wireless Sensor Networks: An Epidemiological Approach

Shashank Awasthi[1,2] · Pramod Kumar Srivastava[3] · Naresh Kumar[2] · Rudra Pratap Ojha[2] · Ashok Kumar Yadav[4]

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Abstract
Epidemic spreading in wireless sensor networks (WSNs) has lately attracted the attention of many researchers as a hot problem in nonlinear systems. Using wireless connectivity, the sensor nodes that makeup WSNs are linked to one another in a decentralized and distributed structure. Decentralized architectures and resource limitations pose a security risk for WSNs. Malware attacks the WSN's sensor nodes, paralyzing them while collecting data from the network. Attacks by malware can increase the energy use of WSN sensor nodes. It just began spreading from one infected node and uses nearby nodes to expand over the whole WSN. Therefore, the protection of WSNs against malware attacks is a need that cannot be avoided. In this research paper, we propose an epidemic model to discuss the impact of charging on sensor nodes, coverage, and connectivity, considering prior research. The suggested model examines the dynamics of malware spread in WSNs and also explains how much energy is used by the sensor node. The stability of the system has been examined in terms of endemic and local equilibriums for malware propagation. The fundamental reproduction number expression, which is used to assess the prevalence of malware in WSN, has been calculated for the analysis of system dynamics. This study explains the implications of communication radius, sensor node charging, node density, and deployment area on malware dissemination. In comparison to previous models, the suggested approach offers a superior method to stop the propagation of malware in WSN. At the end, we show the results of a numerical simulation of how malware spreads, which proves that our theoretical approach is correct.

Keywords Fundamental reproduction number · Equilibrium points · Low energy · Malware · Stability · Wireless sensor network

Introduction

A wireless sensor network (WSN) is a kind of ad-hoc network that can be installed on any type of terrain. This is one of the beautiful advantages of WSN. Along with the advantages, some problems are faced by WSN due to resource restrictions on sensor nodes. With increasing applications of WSN, various types of problems arise, such as security and the lifetime of sensor nodes due to the vulnerable structure of the network and the limitations of the battery. Malware attacks have become prominent in WSN due to these restrictions. When a malware attack happens, information starts to leak out of the network, and more energy is lost than during normal operation. To overcome the issue of limited storage capacity of energy and the troublesome replacement mechanism of batteries, technology has been developed known as wireless rechargeable sensor networks (WRSNs). Denial of Service (DOC) attacks also affect rechargeable sensor nodes, with disastrous effects for real-time and pre-warning applications. Therefore, research on WSN security is crucial and essential. Malware and breaches in information security are currently contributing to a new category of problems for human society. Malware attacks make it difficult for the WSN to carry out its functions. Because of the unique qualities that it possesses, WSN is useful in a diverse set of settings, including those pertaining to agriculture, healthcare, defense, mobility, and many more [1, 2]. When it comes to the Internet of Things (IoT) and the Internet of Vehicles (IoV), sensor nodes play an important role in the transmission of data [3]. Tens of thousands of sensor nodes make up a WSN, and these

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Extended author information available on the last page of the article

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nodes can be dispersed in a manner that is either completely at random or according to a specific pattern [4, 5]. The primary responsibilities of sensor nodes are the collection of environmental data and the transmission of that data to the node that is being targeted (the sink node). The sink is located a significant distance from the sphere of influence of the sensor node, so the information must be transmitted there via a series of hops [6, 7]. The sensor node is a piece of hardware that has limited processing power, a limited energy source, a limited memory capacity, and a limited coverage area. Because of this, WSN has its own set of unique operational problems, such as limited energy, security, and where to put nodes [8, 9]. Malware attacks on WSNs are one of the most vital issues that need to be addressed. Malware that is created purposely and deliberately installed does not normally enter a network. Malware has the ability to activate itself and spread across WSNs through the process of self-transmission [10], so it can do so even in the absence of human interaction. The malicious software has the capability to bring the network to a halt, just as the infamous "beauty killer" worm attack did [11], which wiped out all Internet services. The Chernobyl worm was another virus assault which rendered the computer system inoperable by corrupting the system's vital input and output components through the infections of application programs [12].

A WSN is a network of minimal-power sensor devices that transmit data via wireless communication. This kind of computer network [1] is vulnerable to virus attacks. As compared to computer networks, the topology of WSNs is more complex and prone to malware assault [13]. Inadequate defense mechanisms and a scattered organizational structure make WSN an ideal environment for malware to spread. Because sensor nodes lack adequate defenses, malware can easily attack them. Many researchers have investigated how malware attacks the network and have proposed and created a variety of countermeasure measures to stop malware from spreading on the WSN. The idea of epidemic modeling helps researchers figure out how malware spreads in WSN, because the way malware acts in WSN is similar to how diseases spread in a population. For the investigation of infectious diseases in a population, Kermack and McKendrick [13] introduced the SIR model for the first time in 1927. Using the epidemiological modeling paradigm, researchers investigate the spread of worms or malware in WSN [1], computer networks [14], and the spread of speculation in online social networks [15]. Taking into account a variety of epidemic models, the various authors discuss how worms and viruses spread through WSN and how those models compare to one another. These models focus on the many different aspects of worm or virus transmission. Due to the wide variety of applications for WSNs, the protection of these networks from malicious software has become an important area of research. It is absolutely necessary to have a comprehensive understanding of the dynamics involved in the spread of malware in WSN to design an effective defense mechanism for WSN against the attack of malware. The current body of research provides compelling evidence in support of the hypothesis that epidemic models can be helpful in describing various facets of the spread of malware. Researchers have turned to the theory of epidemic modeling to provide an explanation for the process by which worms spread throughout WSN [1, 16–20].

An epidemic model explicates the dynamics of worm propagation in WSN while also taking into account the multiple stages that make up an epidemic [1]. They also take into consideration how the worm's ability to spread alters over time and across space. They were able to determine the worm's behavior in terms of its ability to spread by employing both mathematical analysis and computer simulations. During the process of analyzing the performance of the model, a number of factors are taken into account. In the paper [16], Upadhyay and Kumari investigated an epidemic model that took into account the combative behavior of worms in WSN. To better illustrate how worms spread throughout a network, the concept of a terminally infected node was developed. This concept is used to refer to a node that is completely infected. A sensor node's rate of energy consumption can be optimized through the utilization of the concept of a sleeping node. Ojha et al. came up with the idea for the SEIRS model. This model additionally elucidates how the WSN worm propagates throughout the network. In addition, Wang and colleagues [17] suggested combining the sleeping and awake modes of sensor nodes to lessen the damage caused by worm attacks and save energy. Keshri and Mishra [18] developed the time delay approach to lessen the amount of damage caused to WSN nodes. Certain researchers took into consideration the energy problem that was present in WSN, and they came up with a variety of theories and approaches to lengthen the lifespan of WSN. The mobile charging strategy was developed by Moo et al. [19] as part of an effort to make wireless sensor nodes survive for a longer period of time. An epidemiological model was proposed by Liu et al. [20], and it was based on the SILRD model. In their model, they fail to account for the essential dynamical component that is built into the theory of epidemics. This component is inherent in the theory. To classify the states of the sensor nodes, it is necessary to take into account the energy level of the nodes. They utilized strategies from game theory to optimize the energy utilization of sensor nodes. Additionally, they stopped the spread of malicious programs' infections throughout the WSN. In addition, the authors did not discuss the fundamental reproduction rate, essential dynamics, or equilibrium positions of the system. The basic reproduction number is one of the essential components of epidemic modeling, which is utilized to characterize the level of infection in a system. This number is one of the key elements. They also make no attempt to rectify the precarious situation of the infrastructure of the WSN in any manner. These are concerns that are of the utmost significance to the system.

To address the challenges that was brought up in our previous work, we have developed a model that describes the dynamics of how malware spreads through WSN. The mechanism was developed for the purpose of preventing attacks caused by malware as well as optimizing the amount of energy consumed by sensor nodes. The concept of an exposed state of epidemiology is utilized in the model that has been proposed to perform early stage detection of the presence of malware in WSN. Additionally, the model investigated the effects that charging would have on the spread of malware. The model also looked at how the number of nodes, the number of nodes in a certain area, and the area where the nodes are placed all affect how malware spreads in WSN.

Major Contribution

This paper addresses these significant issues and discusses them. The following is a summary of the most important contributions made by the proposed models:

- Devices with minimal energy use are the sensor nodes. The energy used by sensor nodes during operation causes them to enter a low-energy condition. The sensor node uses more energy as an outcome of the malware assault. One of the epidemic model's states in the suggested model is the low-energy class of the sensor nodes.
- The charging method is utilized to prevent the spread of malware, improve network security, and maintain the stability of the WSN.
- The epidemiological model that has been recommended looks at how malware spreads in a WSN in a dynamic way and proposes a way to stop it spreading.
- Check out how the system responds when it is attacked by WSN malware, look for points of equilibrium, and talk about how stable the system is in different situations.

Roadmap of Article

The following is an outline of the structure of the remaining portion of the paper: In Sect. "Proposed Model", we will discuss the model that has been proposed. Model description, formulation, and analysis are all broken down and explained in Sect. "Model Description, Formulation, and Analysis". In Sect. "Existence of Equilibrium Point and Its Stability Analysis", we talk about the stability of the system, essential theorems, and justifications for those theorems. The revised SEILRD model is discussed in Sect. "Improved SEILRD Model" of the report. In Sect. "Node Communication Radius and System Performance", we will discuss the impact that the communication radius has on the overall performance of the system. The effect that the number of nodes has on the overall performance of the system is discussed in Sect. "Effects of Node Density on System Performance". In Sect. "Performance Analysis", we go into detail about the performance analysis of the proposed model in comparison to other models that are currently on the market. The conclusions are discussed in Sect. "Conclusion", along with suggestions for further research.

Proposed Model

The Susceptible-Exposed-Infectious-Low Energy-Recovered-Dead (SEILRD) model based on the concept of epidemiology is proposed. The model helps in the study of malware dynamics in WSN. The outcome of a malware attack on WSN is to steal information from the network, consume the energy of sensor nodes, and paralyze network operations. Consequently, to stop the spread of malware in WSN, it is essential to first determine whether or not the network is being attacked by malware. Therefore, for this purpose, the exposed state of the epidemic model is useful, because it helps in finding the malware presence in the network at an early stage. The important contribution of the proposed model is:

1. The model, which is based on the theory of epidemics, has been developed to investigate the dynamics of malware transmission in WSN and to develop controlling mechanisms.
2. The model makes it possible to determine the existence of malware in a WSN at an early stage using the vulnerable state, which is beneficial to the mitigation of malware transmission.
3. Investigate the impact that charging and patching have on the transmission and recovery of malware on the responsiveness of the system.
4. Examine the degree to which the system can remain stable under a wide range of circumstances.
5. Determine the points of equilibrium and conduct a performance evaluation on the various proposed models. Determine the points of equilibrium and conduct a performance evaluation on the various proposed models.
6. The proposed models are proven to be correct both theoretically and with verified results from extensive simulations.

The different states of the proposed models: Susceptible State (S): Sensor nodes that are uninfected by any type of malware and have a high-energy level. They can execute the assigned task in the normal way, but due to a lack of a defense mechanism, they are extremely vulnerable to the attack of malware.

Exposed State (E): Susceptible sensor nodes come into contact with malware converted into an exposed state. Or when susceptible sensor nodes installed with malware become exposed nodes but are not executed completely. And after a time, interval (latent period), they become infectious.

Infectious State (I): Exposed sensor nodes become infected after a latent period if a corrective measure is not applied on time. These sensor nodes have the ability to infect the environment. The energy consumption increases rapidly when the sensor node comes into an infectious state. If these nodes are not charged or installed with anti-malware in time, they will lose their energy and die.

Low-Energy State (L): Sensor nodes lose their energy due to the operation or attack of malware. They do not have sufficient energy to work in a proper manner and cannot transmit data or infection to other sensor nodes.

Recovered State (R): Sensor nodes are equipped with anti-malware and are immune to malware attacks. These nodes have a high-energy level. The anti-malware installation and charging take place simultaneously. The anti-malware can protect sensor nodes against relevant malware attacks. These nodes can lose their energy and protection against inhomogeneous malware attacks.

Dead State (D): Sensor nodes have lost their energy due to the use of energy during normal operation or maybe the failure of hardware or software. These nodes are completely dysfunctional. These sensor nodes cannot be worked on after charging either. They do not have the capability to perform any kind of operation like the collection, processing, and transmission of data.

The development of methods to control the spread of malware and increase the lifetime of WSNs is the primary objective of the approach that has been proposed. The model illustrates the dynamics of the spread of malware in WSN, and the following are major contributions made by the proposed model:

1. The transmission dynamics of WSN malware are investigated using the proposed model. This model suggests defense mechanisms against malicious software.
2. To prevent the spread of malware and reduce the amount of energy used by sensor nodes in WSN, the exposed state concept should be utilized.
3. To incorporate low-energy states into the epidemic model, which is used for network maintenance and charging, extends network lifetime, suppresses malicious activity, and improves WSN security.
4. To analyze the system response due to attack of malware and investigate the method of fast recovery of WSN in steady-state conditions.
5. To investigate the stability of the system under various conditions and validate the analytical study through the results of comprehensive simulations.

Some assumptions are made in the formulation of the model. The sensor nodes are homogeneous and distributed in a field for the collection of information. It is assumed that in the beginning, sensor nodes are in a Susceptible (S) state. They are free from malware. The attacker installs malware in WSN through a sensor node, then the state of the sensor node changes, which is shown in Fig. 1.

Model Description, Formulation, and Analysis

Protecting a WSN from an attack by malware necessitates the utilization of a defensive mechanism. To achieve this goal, the model is proposed, and the conceptualization of this model is shown in Fig. 1. The following presumptions have been made in order for the model to be formulated:

1. In the initial state, sensor nodes are in a state known as Susceptible (S), which means they are vulnerable to malware attacks. If malware is allowed to install itself in a vulnerable sensor node, then that node may eventually become an exposed node. Malware can be executed partially on the exposed node. The rate at which a vulnerable node will transform into an exposed node is β As a result of the typical functioning of vulnerable sensor nodes, some of the nodes lose their energy and transition into a low-energy state with rate ρ. Some of the sensor nodes enter a dead state at a rate of α_3 due to a malfunction in either the hardware or the software or because the battery is depleted. Anti-malware software is installed on some of the vulnerable nodes, which gives those nodes immunity against relevant malware attacks and increases the probability that they may recover ε_1.
2. The irregular behavior that was exhibited by the exposed state (E) of the sensor node in WSN. Malware has been successfully installed on the exposed node, but it has not yet been executed in its entirety. Therefore, to put a stop to the further spread of malware in WSN, it is necessary to promptly apply a corrective measure to the exposed state of the nodes. If that does not happen, these nodes

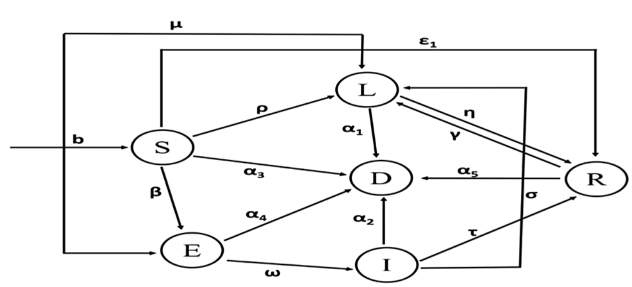


Fig. 1 Transition state diagram of SEIRLD

will quickly become infectious with the rate ω. Some of the exposed nodes may enter in a low-energy state due to loss of energy or contact with malware at the rate \$*mu*; some of the sensor nodes in the exposed class are rendered inoperable at a rate α_4 due to a malfunction in either the hardware or the software, the depletion of the battery, or an attack by malware.

3. The sensor node of the WSN is infected with malware, which may cause malicious activity throughout the network. Infected nodes might begin to spread malware to the nodes that are nearby, which can also result in an increase in the amount of energy that is lost by the infected nodes. These infected nodes are only infectious and not destructive, and malicious programs may decide not to continue attacking the node that has been infected with the virus. If the corrective measure is applied on time and anti-malware can be successfully installed in this manner, the node will be converted back to its recovered state in a safe manner at the rate τ. If malicious programs cause more damage, the node is going to fade away more quickly. Some of the infected sensor nodes may eventually stop functioning due to a problem with either the hardware or the software, because the battery can run out, or because malware can attack them. Infectious nodes loss their energy due to continuous attack of malware and enter into low-energy state (L) with rate σ.
4. Not only are the recovered nodes in a WSN network impervious to attacks from malicious software, but they also have a high level of available energy. With a level of high energy and anti-malware installed in these nodes that do not require charging, sensor nodes of the network transition into a recovered state, having previously been in a vulnerable and infectious state. On the other hand, the low-energy nodes recharge and installation of anti-malware take place at the same time. The changing rate of low-energy state to recovered state is η. Though, nodes turn into low-energy state from recovered state with rate γ due to normal consumption without energy replenishment. Some of the recovered class of sensor nodes enter into dead state with rate α_5 because of hardware/softwaremalfunction or attack of malware.
5. Some of the nodes in the low-energy state may be in a protective state, but not all of them. Rapid depletion can be observed in the low-energy state nodes. As a result, they reach the state of being dead relatively quickly, at the rate of α_1.
6. The system may experience a growth in the number of sensor nodes that are vulnerable to attack at the rate denoted by b

There are six possible states that can be assigned to the overall number of sensor nodes that are part of a WSN at any given time t. Let assume that N(t) indicates number of nodes in the system. All the six classes nodes can be named as Susceptible State S(t), Exposed State E(t)); Low-Energy State L(t); Infectious State I(t)Recovered State R(t) and Dead State D(t). To provide a mathematical representation of the connection that exists between the nodes, a possible approach is to employ the equation that has been stipulated below:

$$N(t) = S(t) + E(t) + I(t) + L(t) + R(t) + D(t), \forall t \geq 0.$$

A visual representation of the transmission dynamics of the SEILRD model is shown in Fig. 1. A set of differential equations has been provided in this article for the purpose of researching the transmission dynamics of malware as well as the state of change of the sensor node. The equations that describe the transitions from one state to another are given as follows. The definitions of the various notations that are used in the equation are provided in Table 1

$$\begin{aligned}
\dot{S} &= b - \beta SI - (\rho + \alpha_3 + \epsilon)S \\
\dot{E} &= \beta SI - (\mu + \omega + \alpha_4)E \\
\dot{I} &= \omega E - (\tau + \sigma + \alpha_2)I \\
\dot{L} &= \mu E + \sigma I + \gamma R - (\alpha_1 + \eta)L + \rho S \\
\dot{R} &= \tau I + \epsilon S + \eta L - (\gamma + \alpha_5)R \\
\dot{D} &= \alpha_1 L + \alpha_2 I + \alpha_3 S + \alpha_4 E + \alpha_5 R.
\end{aligned} \tag{1}$$

Existence of Equilibrium Point and Its Stability Analysis

Existence of Malware-Free Equilibrium

The first five equations of Eq. (1) are independent of the sixth equation, since it is obvious from Eq. (1) that nodes of class D are absent from the first five equations. Therefore, equate to zero all first-order derivatives of Eq. (1) to establish the system's equilibrium points

$$\begin{aligned}
0 &= b - \beta SI - (\rho + \alpha_3 + \epsilon)S, \\
0 &= \beta SI - (\mu + \omega + \alpha_4)E, \\
0 &= \omega E - (\tau + \sigma + \alpha_2)I, \\
0 &= \mu E + \sigma I + \gamma R - (\alpha_1 + \eta)L + \rho S, \\
0 &= \tau I + \epsilon S + \eta L - (\gamma + \alpha_5)R.
\end{aligned} \tag{2}$$

By solving Eq. (2), we can obtain information regarding the points at which the system is in a state of equilibrium. The point at which this occurs is referred to as the malware-free equilibrium (MFE), and it is indicated by $P_0^{MFE} = \left(S_0, E_0, I_0, L_{0,} R_0\right)$ where $S_0 = \frac{b}{(\alpha_3+\rho+\epsilon)}$, $E_0 = 0, I_0 = 0,\ L_0 = \left\{\frac{\rho(\gamma+\alpha_5)+\epsilon\gamma}{(\alpha_1\alpha_5+\eta\alpha_5+\alpha_1\gamma)}\right\}S_0$, $R_0 = \left\{\frac{\rho\eta+\alpha_1\epsilon+\epsilon\eta}{(\alpha_1\alpha_5+\eta\alpha_5+\alpha_1\gamma)}\right\}S_0$.

Table 1 Used parameters and their meanings

S.No	Parameter	Meaning of the used parameter
1	b	Network node vulnerability rate
2	α_1	Low-energy nodes enter dead nodes
3	α_2	Rate of malware-infected nodes drain energy and die
4	α_3	Rate of vulnerable nodes becoming dead nodes
5	α_4	The rate of exposed states becoming dead nodes
6	α_5	Rate of dead nodes recovered
7	σ	Infectious state nodes joining low-energy nodes
8	X	Rate of low-energy node recovery
9	μ	Probability of exposed to low-energy nodes
10	ω	Probability of exposed to infectious node
11	ϵ	Probability of Anti-malware installation in susceptible nodes
12	ρ	Probability of high-energy susceptible nodes becoming low-energy
13	β	Probability of malware infection rate of susceptible nodes
14	τ	High-energy infectious node recovery rates
15	Y	Probability of recovering (high energy) to low energy

Existence and Uniqueness of Malware Endemic Equilibrium

In this part of the article, we analyze the presence of the endemic equilibrium as well as its uniqueness. As we know that for malware endemic equilibrium (MEE) point ($E^* = I^* \neq 0$) Therefore, from Eq. (2), we get

$$\begin{aligned} 0 &= b - \beta S^* I^* - (\rho + \alpha_3 + \epsilon)S^*, \\ 0 &= \beta S^* I^* - (\mu + \omega + \alpha_4)E^*, \\ 0 &= \omega E^* - (\tau + \sigma + \alpha_2)I^*, \\ 0 &= \mu E^* + \sigma I^* + \gamma R^* - (\alpha_1 + \eta)L^* + \rho S^*, \\ 0 &= \tau I^* + \epsilon S^* + \eta L^* - (\gamma + \alpha_5)R^*. \end{aligned} \tag{3}$$

By simple calculation, MEE point P^{*MEE} is given by $S^* = \frac{1}{R_0^{th}}$, $I^* = \frac{(bR_0^{th} + \alpha_3 + \rho + \varepsilon)}{\beta}$ $E^* = \frac{(bR_0^{th} + \alpha_3 + \rho + \varepsilon)}{R_0^{th}(\mu + \omega + \alpha_4)}$, $R^* = \frac{(\tau I^* + \varepsilon S^* + \eta L^*)}{(\gamma + \alpha_5)}$, $L^* = \frac{(P_1 + P_2 + P_3)}{(\gamma\alpha_1 + \alpha_5\alpha_1 + \eta\alpha_5)}$, where $P_1 = \frac{((bR_0^{th} + \alpha_3 + \rho + \varepsilon) + \sigma(\gamma + \alpha_5) + \tau\gamma)}{\varepsilon}$, $P_2 = \mu b R_0^{th} + \alpha_3 + \rho + R_0^{th}(\mu + \omega + \alpha_4)$, $P_3 = \rho\gamma + \alpha_5 + \eta\gamma R_0^{th}$, where $R_0^{th} = \frac{\beta b \omega}{(\omega + \mu + + \alpha_4)(\tau + \sigma + \alpha_2)(\rho + \varepsilon + \alpha_3)}$ is basic reproduction number [21]. The existence and uniqueness of the MEE point become evident when $R_0^{th} > 1$. There are two distinct types of system stability, both of which are discussed in this section. We performed an analysis of the system's stability, looking for malware-free equilibrium as well as endemic equilibrium. The theorems and the proofs of those theorems have been provided in this instance with the intention of investigating the system's stability.

Assessment of Local and Global Stability of Malware-Free Equilibrium

The analysis of how a system behaves when it is under assault by malware is one of the most pressing concerns. To investigate whether or not the system is stable, several theorems have been developed. The following theorems will be used to make a decision on the network's stability.

Theorem 1 (Theorem 1) *If $R_0^{th} < 1$, the model depicted in Eq. (2) is said to be in a state of locally asymptotically stable at Malware-free equilibrium (MFE), but if $R_0^{th} > 1$, the system is said to be unstable.*

Proof By forming a Jacobian matrix, which aids in eigenvalue determination, it is able to determine whether or not the system at point P is stable. As a result, the equivalent Jacobian matrix is

$$(P_0^{MFE}) = \begin{pmatrix} -(\rho+\varepsilon+\alpha_3) & 0 & -\beta S_0 & 0 & 0 \\ 0 & -(\mu+\omega+\alpha_4) & \beta S_0 & 0 & 0 \\ 0 & \omega & -(\tau+\sigma+\alpha_2) & 0 & 0 \\ \rho & \mu & \sigma & -(\alpha_1+\eta) & \gamma \\ \varepsilon & 0 & \tau & \eta(\alpha_5+\gamma) & \end{pmatrix}. \quad (4)$$

Three eigen values of Eq. (4) are: $\kappa_1 = -(\rho+\alpha_3+\varepsilon), \kappa_2 = -(\mu+\alpha_4+\omega), \kappa_3 = -(\tau+\alpha_2+\sigma)$ eigen values are the roots of equation $a_0\kappa^2 + a_1\kappa + a_2 = 0$, where $a_0 = (1-R_0^{th}),\ a_1 = (\alpha_1+\eta+\gamma+\alpha_5)(1-R_0^{th}),\ a_2 = (\alpha_1+\eta)(\gamma+\alpha_5) - R_0^{th}(\alpha_1(\gamma+\alpha_5)+\eta\alpha_5)$ when $R_0^{th} < 1$, all of the coefficients a_0, a_1 and $a_2 = 0$ positive. Therefore, it is clear that none of the matrix's eigen values are positive when $R_0^{th} < 1$. Therefore, in a state of equilibrium where there is no malware, the system is locally asymptotically stable at P_0^{MFE} and when $R_0^{th} > 1$, system is unstable. □

Theorem 2 *Theorem 2 If basic reproductive number is less than or equal to one i.e.,$R_0^{th} \le 1$, the Malware-Free Equilibrium (MFE) P_0^{MFE} is said to be globally asymptotically stable.*

Proof Proof. Consider the Lyapunov function as follows:

$$L(t) : \mathbb{R}^5 \to \mathbb{R}^+ definedby L(t) = \zeta_1 E + \zeta_2 I. \quad (5)$$

Now, taking the first derivative of (5), we get

$$\dot{L} = \zeta_1\dot{E} + \zeta_2\dot{I} = \zeta_1\big(\beta SI - (\mu+\omega+\alpha_4)E\big) + \zeta_2\big(\omega E - (\tau+\sigma+\alpha_2)I\big) \le \big(R_0^{th} - 1\big)I,$$

where $\zeta_1 = \omega, \zeta_2 = (\omega+\mu+\alpha_4)$ if $R_0^{th} \le 1$, then $\dot{L}$ holds. Moreover, $\dot{L} \le 0$ if and only if $I = 0$. Thus, the largest invariant set in $S, E, I, L, R \in \Gamma^{\}} : L \le 0$ is the singleton set. Hence, the global stability of P_0^{MFE} when $R_0^{th} \le 1$ follows from LaSalle's [22] invariance principle. □

To validate Theorems 1 and 2 experimentally, a simulation is carried out, so that the theoretical findings can be evaluated. The findings of the simulation provide evidence that the theory is appropriate. The effect that a number of different parameters have on the propagation of malware in WSN has been investigated. MATLAB (R2018a) is utilized for the purpose of simulations. For computation, taking the value of various parameters are $b = 0.2, \beta = 0.001, \alpha_1 = 0.00012, \alpha_2 = 0.0015,$ $\alpha_3 = 0.00001, \alpha_4 = 0.00011, \alpha_5 = 0.00013, \rho = 0.002,$ $\omega = 0.001, \gamma = 0.004, \sigma = 0.005, \varepsilon = 0.0002, \tau = 0.003, \eta = 0.004.$ Assume that the number of nodes in the various state at time $t = 0 is S(0), E(0), I(0), L(0), R(0)$ and $D(0)$ be 990, 9, 1, 0, 0 and 0, respectively.

Figure 2 shows the changes in number of sensor nodes in different states with respect to time when $R_0^{th} > 1$. The computed value of $R_0^{th} = 3.063047$ which is greater than 1 that means malware persist in WSN. Malware transmits in the network continuously. From Fig. 2, it is noticed that in the beginning, the number of susceptible nodes is decreasing with time, but others are increasing. It has been noticed that the number of dead nodes is increasing linearly with time. The reason for increasing the number of dead nodes is exhaustion of their energy or failure of hardware or software. Figure 2 satisfies the second condition of Theorem 1 when $R_0^{th} > 1$ system will be unstable.

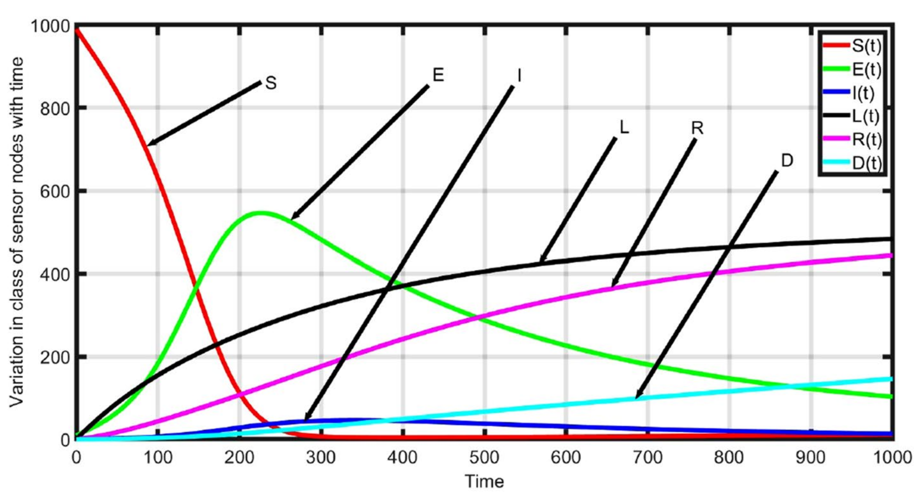


Fig. 2 Transmission dynamics of malware when $R_0^{th} > 1$

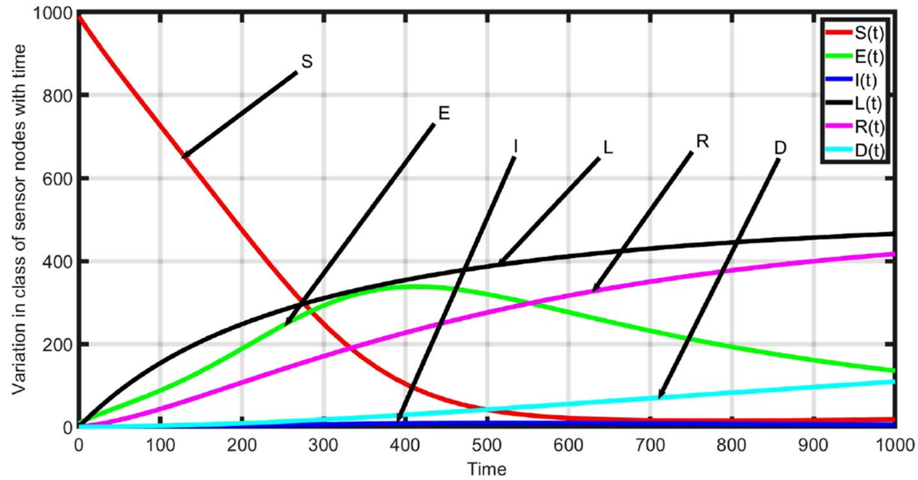


Fig. 3 Transmission dynamics of malware when $R_0^{th} < 1$

When vary some parameters such as $b = 0.15, \omega = 0.0003$, the value of R_0^{th} also vary. Now, the commuted value of $R_0^{th} = 0.889364$, which is less than 1. Figure 3 shows the changes in number of sensor nodes in different states in respect to time when $R_0^{th} < 1$. From Fig. 3, it is observed that a number of susceptible nodes are decreasing in the beginning and other state of nodes are increasing. The infectious and exposed class of nodes reached the maximum value, and then started to decrease and become zero with time. This indicates that malware will not exist in the network for a long time or will be exterminated from WSN. Figure 3 satisfies the first condition of Theorem 1, when $R_0^{th} < 1$ system will be stable and analytical study verified by Theorems 1 and 2. Therefore, the system is stable both locally and asymptotically in its equilibrium free of malicious software. Coverage and connectivity are two of the parameters that need to be taken into consideration during the design phase of the network. To enhance the performance of the model, it is necessary to incorporate the idea of communication radius.

Improved SEILRD Model

Modifications are made to the SEIRLD model to investigate how coverage and connectivity in the WSN influence the spread of malicious software. In this particular model, the sensor nodes are standardized and distributed in an even manner across a two-dimensional space that is measured in length l. Therefore, the total area (A) in which the sensor nodes are deployed is l^2. The total N number of sensor nodes is deployed in the area of $A = L^2$. Therefore, the average node density is $d = \frac{N}{A}$. The communication radius of each sensor node which is deployed in area of A is r. Hence, the coverage area of each sensor node will be πr^2. The coverage area of sensor node indicates that a sensor node can communicate to all nodes which are lying in their coverage area. In WSN, the sensor node detects the events in their surrounding area and collects the event data first after that transmits to the destination node (sink node) in one hop or multiple hops. We assume that at the beginning, all of the sensor nodes are in a vulnerable state and could be the target of a malware attack. In this study, it is assumed that at any time t, total number of N(t) sensor nodes are distributed in the area of l^2 and the number of susceptible sensor nodes are of S(t). A sensor node having the communication radius r and its coverage area is πr^2. Therefore, in per unit area density of susceptible nodes

$$d(t) = \frac{S(t)}{A}. \tag{6}$$

Coverage area of a sensor node

$$A_r = \pi r^2. \tag{7}$$

The total number of sensor nodes that are situated within the range of a sensor node that is vulnerable to being compromised

$$S'(t) = A_r d(t). \tag{8}$$

From Eqs. (6) and (7) putting the value of $d(t)$ and $A_r$in Eq. (8), we get

$$S'(t) = \pi r^2 \frac{S(t)}{l^2}. \tag{9}$$

Therefore, as per relationship of state transition in Fig. 1, the mathematical model of malware transmission in WSN can be derivative as follows:

$$\begin{aligned}
\dot{S} &= b - \xi SI - \rho + \alpha_3 + \epsilon S, \\
\dot{E} &= \xi SI - \mu + \omega + \alpha_4 E, \\
I &= \omega E - \tau + \sigma + \alpha_2 I, \\
L &= \mu E + \sigma I + \gamma R - \alpha_1 + \eta L + \rho S, \\
R &= \tau I + \epsilon S + \eta L - \gamma + \alpha_5 R, \\
D &= \alpha_1 L + \alpha_2 I + \alpha_3 S + \alpha_4 E + \alpha_5 R.
\end{aligned} \tag{10}$$

Existence of Malware-Free Equilibrium and Endemic Equilibrium

The class D is not a part of the first five equations of Eq. (10). Therefore, the first five equations of Eq. (10) are not dependent on the sixth equation. Hence, to determine the system's equilibrium points, equate the first derivative of Eq. (10) to zero

$$\begin{aligned}
&0 = b - \xi SI - \rho + \alpha_3 + \epsilon S,\\
&0 = \xi SI - \mu + \omega + \alpha_4 E,\\
&0 = \omega E - \tau + \sigma + \alpha_2 I,\\
&0 = \mu E + \sigma I + \gamma R - \alpha_1 + \eta L + \rho S,\\
&0 = \tau I + \epsilon S + \eta L - \gamma + \alpha_5 R.
\end{aligned} \tag{11}$$

Solving Eq. (11) yields system equilibrium points. The MFE point is: $P_0^{MFE} = \left(S_0,\ E_0,\ I_0,\ L_{0,}\ R_0\right)$, where $S_0 = \frac{b}{(\alpha_3+\rho+\varepsilon)}, E_0 = 0, I_0 = 0,\ L_0 = \left\{ \frac{\rho(\gamma+\alpha_5)+\varepsilon\gamma}{(\alpha_1\alpha_5+\eta\alpha_5+\alpha_1\gamma)} \right\} S_0,\ R_0 = \left\{ \frac{\rho\eta+\alpha_1\varepsilon+\varepsilon\eta}{(\alpha_1\alpha_5+\eta\alpha_5+\alpha_1\gamma)} \right\} S_0$ To investigate existence and uniqueness of MEE point, we have used straight-forward computation of MEE P^{MEE} is given by simple calculation MEE point P^{*MEE} is given by

$$S^* = \frac{1}{R_0^{th}},\ I^* = \frac{(bR_0^{th} + \alpha_3 + \rho + \varepsilon)}{\xi},$$

$$E^* = \frac{(bR_0^{th} + \alpha_3 + \rho + \varepsilon)}{R_0^{th}(\mu + \omega + \alpha_4)}, R^* = \frac{(\tau I^* + \varepsilon S^* + \eta L^*)}{(\gamma + \alpha_5)}$$

$$L^* = \frac{(P_1 + P_2 + P_3)}{(\gamma\alpha_1 + \alpha_5\alpha_1 + \eta\alpha_5)},$$

where

$$P_1 = \frac{\left(\left(bR_0^{th} + \alpha_3 + \rho + \varepsilon\right) + \sigma\left(\gamma + \alpha_5\right) + \tau\gamma\right)}{\varepsilon}$$

$$P_2 = \mu bR_0^{th} + \alpha_3 + \rho + \epsilon R_0^{th}(\mu + \omega + \alpha_4),$$

$$P_3 = \frac{\rho\left(\gamma + \alpha_5\right) + \eta\gamma}{R_0^{th}},$$

where $R_0^{th} = \frac{\xi b\omega}{(\omega+\mu++\alpha_4)(\tau+\sigma+\alpha_2)(\rho+\varepsilon+\alpha_3)}$ is basic reproduction number [21]. The existence and uniqueness of the MEE point become evident when $R_0^{th} > 1$.

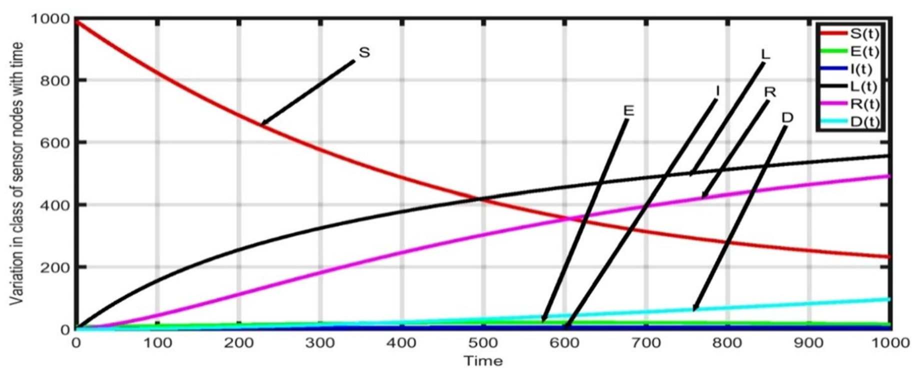


Fig. 4 Malware transmission dynamics in WSN (r = 0.76)

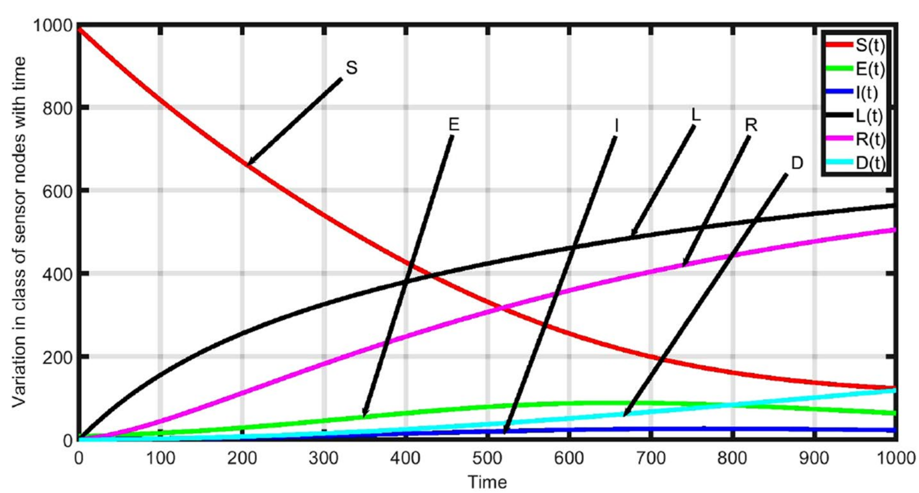


Fig. 5 Malware transmission dynamics in WSN (r = 1.0)

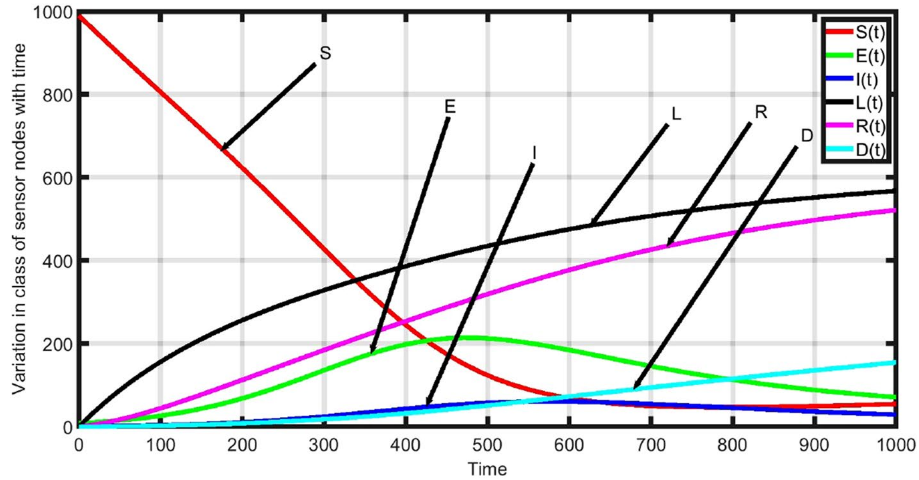


Fig. 6 Malware transmission dynamics in WSN (r = 1.3)

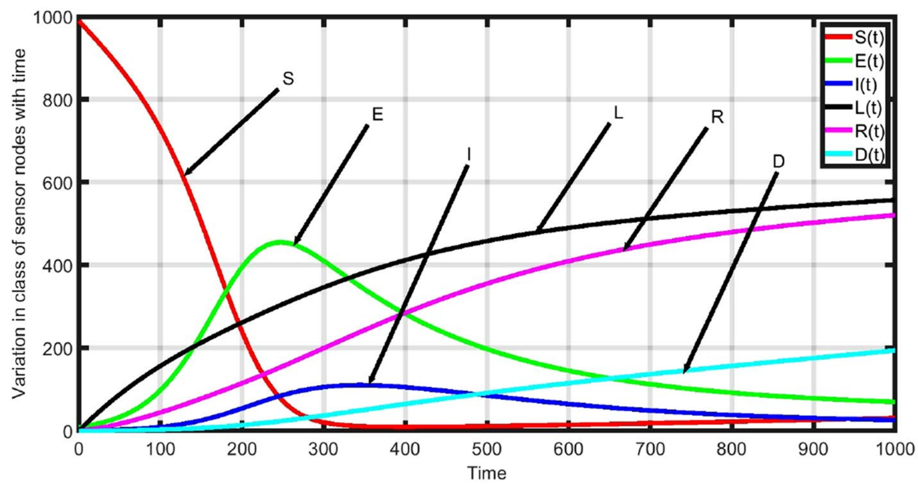


Fig. 7 Malware transmission dynamics in WSN (r = 2.1)

Node Communication Radius and System Performance

We know that

$$R_0^{th} = \frac{\omega\pi b r^2 \beta}{(\omega + \mu + \alpha_4)(\tau + \sigma + \alpha_2)(\rho + \varepsilon + \alpha_3)l^2}. \tag{12}$$

The threshold value of $R_0^{th} = 1$. Therefore, to find the threshold value of communication radius r_{th}, putting the value of $R_0^{th} = 1$ in Eq. (12)

$$r_{th} = l\left(\frac{(\omega + \mu + \alpha_4)(\tau + \sigma + \alpha_2)(\rho + \varepsilon + \alpha_3)}{\omega\pi b\beta}\right)^{\frac{1}{2}}. \tag{13}$$

For simulation, the following parametric values are taking into account: $b = 0.4, \beta = 0.005, \alpha_1 = 0.00012, \alpha_2 = 0.0015,$,

$\alpha_3 = 0.00001, \alpha_4 = 0.00011, \alpha_5 = 0.00013, \rho = 0.002,$

$b = 0.4, \beta = 0.005, \alpha_1 = 0.00012, \alpha_2 = 0.0015,$

$\alpha_3 = 0.00001, \alpha_4 = 0.00011, \alpha_5 = 0.00013, \rho = 0.002,$

$\mu = 0.002, \omega = 0.003, \gamma = 0.004, \sigma = 0.005,$

$\varepsilon = 0.0002, \tau = 0.003, \eta = 0.004, l = 14.$ Putting these parametric values in Eq. (12) and calculated value of $r_{th=}$1.056467. Assume that for the different value of number of nodes in the various state at time t=0 is S(0), E(0), I(0), L(0), R(0), and D(0) be 990, 9, 1, 0, 0, and 0, respectively. The simulation results are illustrated in the Figs. 4, 5, 6, 7.

Case 1: If $r \le r_{th}$, then $R_0^{th} \le 1$; in this scenario, the system is free of malware, and even if malware does exist, it will become extinct within the system and reach a stable equilibrium point where it is malware-free. The evidence for this can be found in Theorem 2.

Figures 4, 5 show the system stability in the malware-free state and in this case value of $r \le r_{th}$ and $R_0^{th} \le 1$. For Fig. 4, the value of r = 0.76 and then $R_0^{th} = 0.517506$, and for Fig. 5, the value of r = 1 and then $R_0^{th} = 0.895959$. Both will satisfy the condition of malware-free equilibrium and the system exists in a stable state. The simulation outcomes are consistent with theoretical findings. Figures 4,5 justify the theoretical findings.

Case 2: If $r > r_{th}$, then $R_0^{th} > 1$, in this case, the system is in an unstable state and malware will exist in the system. This is supported by Theorem 1 of the second condition. And in this case, the system will be in a stable state at endemic equilibrium.

Figures 6, 7 show the system stability at endemic state and in this case value of $r > r_{th}$ and $R_0^{th} > 1$. For Fig. 6, the value of r = 1.3 and then $R_0^{th} = 1.514170$, and for Fig. 7, the value of r = 2.1 and then $R_0^{th} = 3.951178$. Figs. 6, 7 satisfy the condition of endemic equilibrium and system in the unstable state. Figures 6, 7show that in the beginning susceptible number of nodes are decreasing and others are increasing. The number of exposed and infectious nodes reaches to the maximum value after that start to decrease and attain the steady state and on other side then susceptible number node starts to increase. This shows that malware exist continuously in the system. The simulation outcomes are consistent with theoretical findings. The number of dead nodes linearly increasing in the system due to drain of sensor nodes energy or failure of software/hardware. Connectivity establishment among nodes of WSN is a crucial problem; the important factor which influences the connectivity of network is communication radius (r). Therefore, connectivity is a function of r. On the basis of above study based on communication radius (r), the following observations are highlighted.

1. For improvement in the connectivity of network, it is required to increase the size of communication radius (r). With the increase in the size of value of r, R_0^{th} start to increase. The system will be stable in malware-free state when value of $r \le r_{th}$ which explanation has been discussed in case 1.
2. The infectious number of nodes increases with increase in size of r, because if size of r is large, then the greater number of susceptible sensor nodes come in contact with an infectious node simultaneously. Therefore, in this condition, a greater number of susceptible nodes are converted into an infectious state at a time. The details are discussed in case 1 and case 2.
3. The impact of r on the transmission dynamic of malware shows that the threshold value of r optimizes that help in controlling the malware transmission, malware eradication, and enhancement in lifetime of WSN.
4. Basic reproduction number R_0^{th} is directly proportional to communication radius (r).

Effects of Node Density on System Performance

Putting the value of $R_0^{th} = 1$ in Eq. (12), the threshold value of node density d_{th} is determined as follows:

$$1 = d_{th} \frac{\omega \pi b r^2 \beta}{N(\omega + \mu + \alpha_4)(\tau + \sigma + \alpha_2)(\rho + \varepsilon + \alpha_3)}$$

$$d_{th} = \frac{N\left(\omega + \mu + \alpha_4\right)\left(\tau + \sigma + \alpha_2\right)\left(\rho + \varepsilon + \alpha_3\right)}{\pi r^2 b \omega \beta}. \quad (14)$$

For simulation, the following parametric values are taking into account:

$$b = 0.4, \beta = 0.005, \alpha_1 = 0.00012, \alpha_2 = 0.0015,$$

$$\alpha_3 = 0.00001, \alpha_4 = 0.00011, \alpha_5 = 0.00013, \rho = 0.002,$$

$$\mu = 0.002, \omega = 0.003, \gamma = 0.004, \sigma = 0.005,$$

$$\mu = 0.002, \omega = 0.003, \gamma = 0.004, \sigma = 0.005,$$

$\varepsilon = 0.0002, \tau = 0.003, \eta = 0.004, r = 1.0$ Putting these parametric values in Eq. (14) and calculated value of $d_{th} = 5.694504$. Assume that for the time t=0, the number of nodes in the various state is S (0), E (0), I (0), L (0), R (0), and D (0) be 990, 9, 1, 0, 0, and 0, respectively. The simulation results are illustrated in Figs. 8, 9, 10, 11.

Case 1: If $d \le d_{th}$, then $R_0^{th} \le 1$; in this scenario, the system is free of malware, and regardless of whether malware does exist, it will eventually become extinct in the system and stabilize at a state where it does not contain any malware. The aforementioned claim is validated by Theorem 2.

Figures 8, 9 show the system stability in malware-free state and in this case value of $d \le d_{th} and R_0^{th} \le 1$. For Fig. 8, the value of d=3.2, l=17.67767, and then, $R_0^{th} = 0.561945$, and for Figure 9, the value of d=4.9, l=14.285714 and then $R_0^{th} = 0.860479$. Figures 8, 9 justify the theoretical findings. In the beginning, susceptible number of sensor nodes is decreasing and remaining are increasing. After some time, infectious number of nodes becomes zero in the system. This can be shown in simulation by increasing the time. Figures 8, 9 satisfy the condition of malware-free equilibrium and system will exist in stable state. The consumption of energy by sensor nodes or the malfunction of either software or hardware is leading to an exponential rise in the number of dead nodes in the system. Case 2: If $d > d_{th}, then R_0^{th} > 1$; in this case, the system is in the unstable state and malware will exist in the system. This is supported by Theorem 1 of

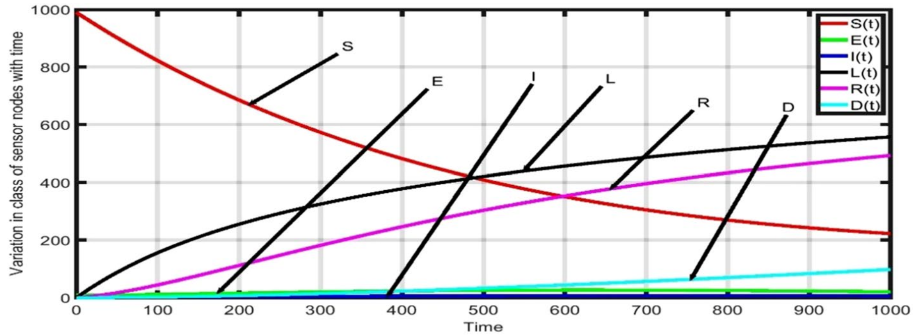


Fig. 8 Malware transmission dynamics in WSN (d = 3.2)

second condition. And in this case, system will be in stable state at endemic equilibrium.

Figures 10, 11 show the system stability at endemic state, and in this case, value of $d > d_{th} and R_0^{th} > 1$. For Fig. 10, the value of d=6.3, l=12.598816, then $R_0^{th} = 1.10633$, and for Fig. 11, the value of d=11, l=9.534629, and then, $R_0^{th} = 1.931687$. Figures 10, 11 satisfy the condition of endemic equilibrium and the system in an unstable state. Figures 10, 11 show that in the beginning, susceptible number of nodes is decreasing and others are increasing. The number of exposed and infectious nodes reaches to the maximum value after that start to decrease and attain a steady state and on other side then susceptible number node starts to increase. This shows that malware exists continuously in the system. The simulation outcomes are consistent with theoretical findings. The energy of sensor nodes is being consumed at a linearly increasing rate, which is leading to an increase in the number of dead nodes in the system. This could also be the result of a failure in either the software or the hardware. Node density d also affects the connectivity of WSN like communication radius r. On the

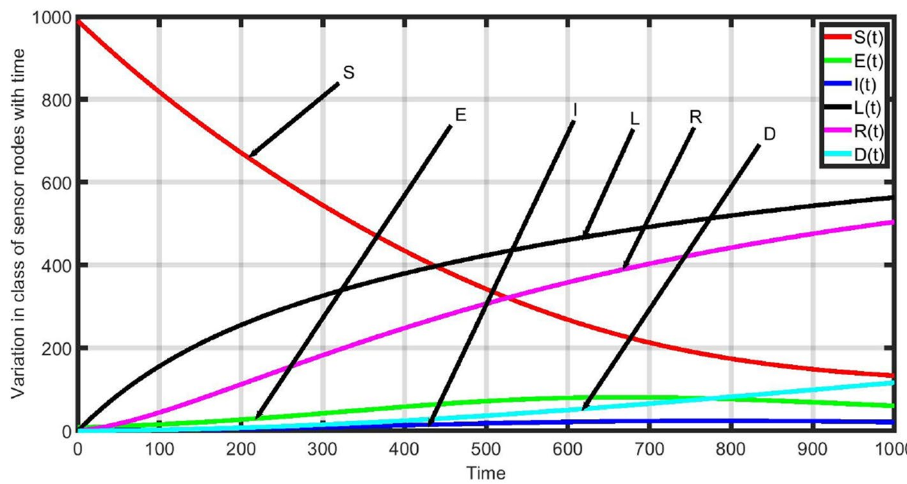


Fig. 9 Malware transmission dynamics in WSN (d = 4.9)

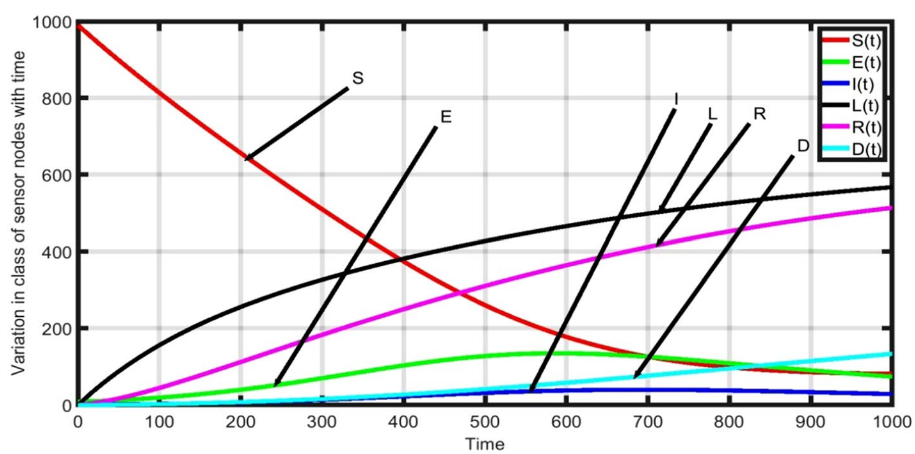


Fig. 10 Malware transmission dynamics in WSN (d = 6.3)

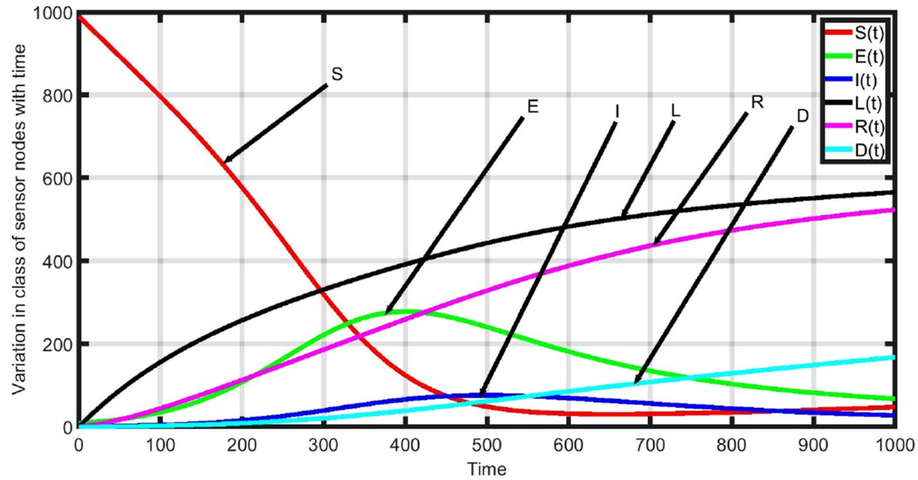


Fig. 11 Malware transmission dynamics in WSN (d = 11)

basis of above study based on node density d, the following observations are highlighted.

1. It can be established from Figs. 8, 9, 10, 11, when increasing in node density, the connectivity among the nodes also increases. When the number of sensor nodes increases and area of deployment is constant, then the distance between sensor will be small. Hence, connectivity among the sensor nodes will be strong. This is explained in case 1 and case 2.
2. As the value of node density increases, the value of R_0^{th} also increases. This weakens the system stability. In the case of high node density, an infectious sensor node can communicate with large number of susceptible sensor nodes simultaneously and infect them. Hence, the value of R_0^{th} increases rapidly due to large number of infectious nodes increase quickly. This can be visualized from Figs. 8, 9, 10, 11.
3. To enhance the lifetime of WSN method of optimization can be applied (by cautiously deployment of sen-

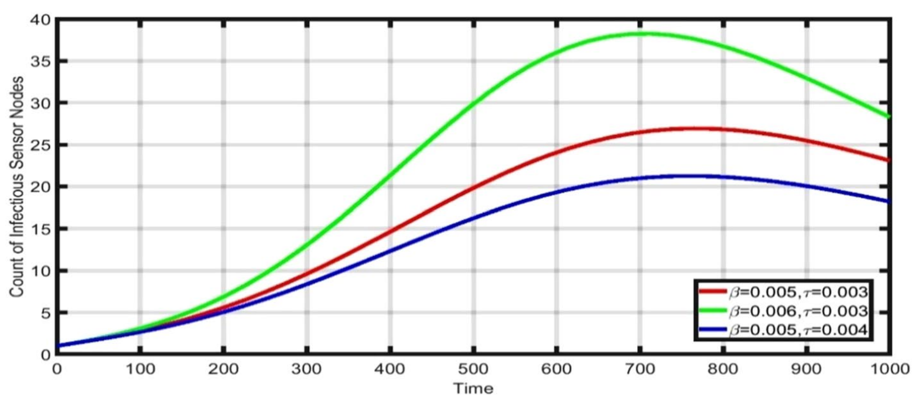


Fig. 12 Impact of recovery rate on infectious nodes

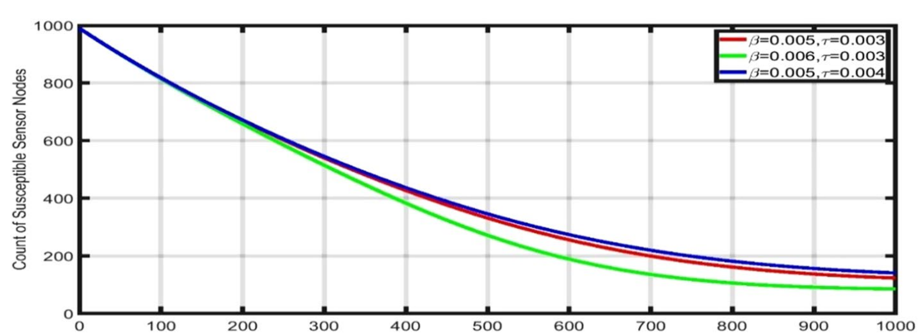


Fig. 13 Impact of recovery rate on susceptible nodes

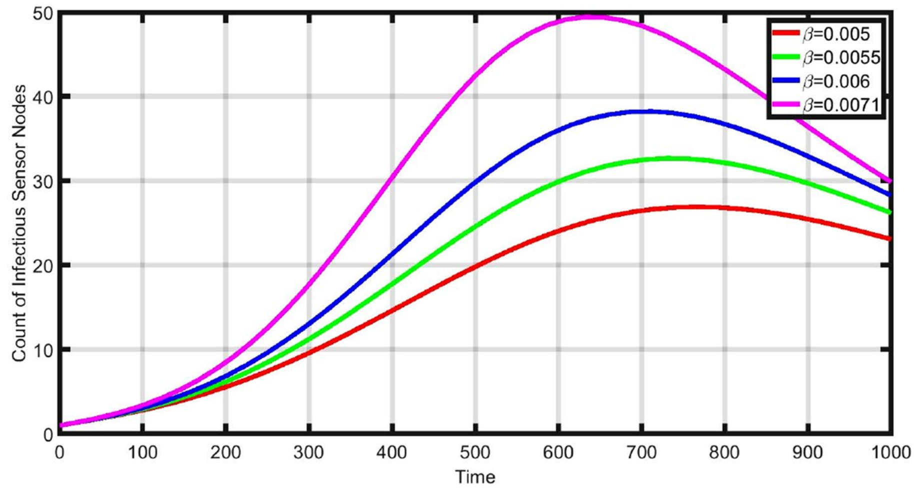


Fig. 14 Impact of infection rate on malware transmission

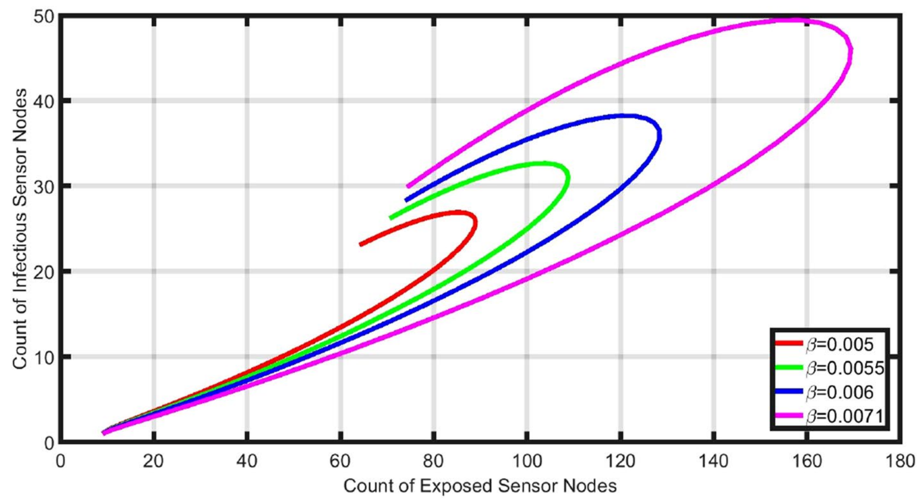


Fig. 15 Impact of recovery rate on susceptible nodes

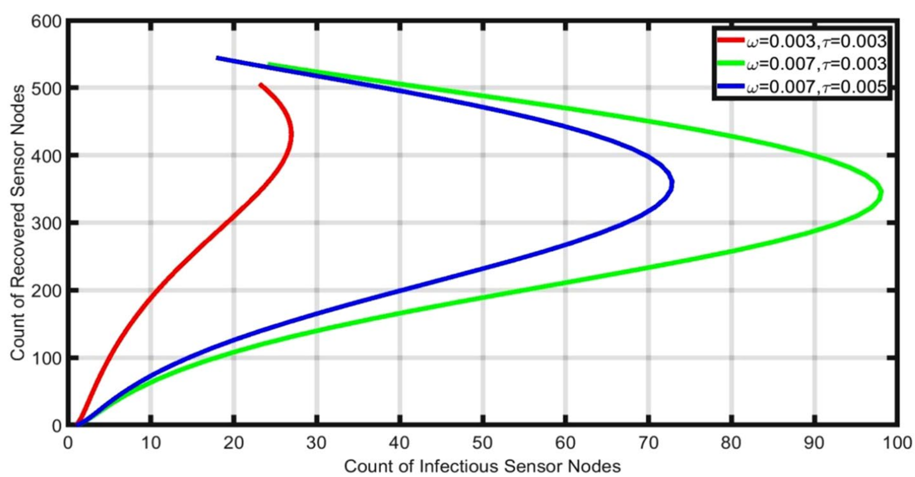


Fig. 16 Impact of recovery rate on exposed nodes

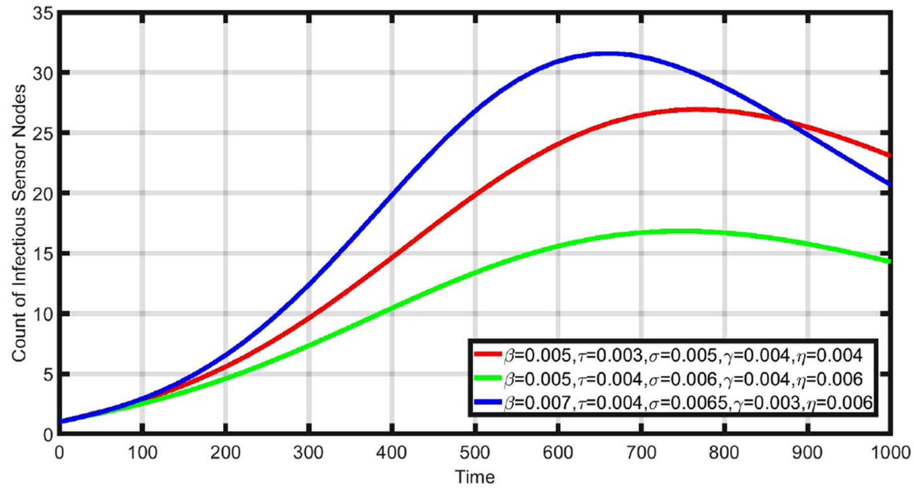


Fig. 17 Impact of various parameters on recovered nodes

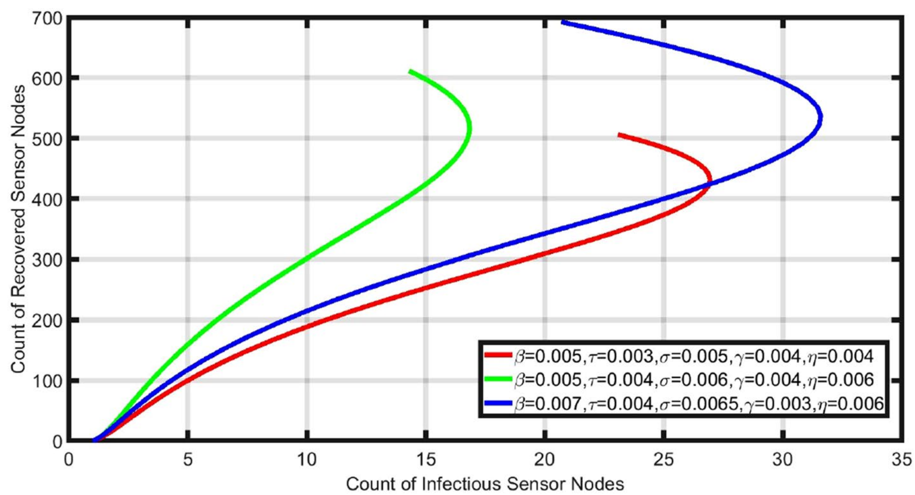


Fig. 18 Impact of various parameters on infectious and recovered nodes

sor nodes in the sensor field). By the use of concept of threshold of node density, we can establish the malware-free network and overcome of communication overhead in the network and made economically viable.

Performance Analysis

The model performance may be impacted by the different criteria, such as rate of infection (β), charging rate (η), recovery rate (τ), and response of susceptible state of sensor nodes. The impact of parameters is elaborated through numerous graphs which are demonstrated in Figs. 12, 13, 14, 15, 16, 17,18

It is observed from Fig. 12 that if value of (τ) is fix and change in value of (β), the count of infectious sensor nodes changes. When increases in the value of (β), the count of infectious sensor nodes may increase in system. Whereas if value of (β) fix and increase in value of (τ), the count of infectious sensor nodes decreases in the system with increase value of (τ). Therefore, to overcome the problem of malware transmission in the network, it is necessary to execute the anti-malware at faster rate in WSN.

The observations of Fig. 12 supported by the Fig. 13. From Fig. 13, it is noted that when the value of (β) is fixed and increase value of (τ), a greater number of susceptible sensor nodes exist in the system with higher recovery rate.

On the other hand, if value of (τ) is fixed and (β) changes lesser number of susceptible sensor nodes exist in the system in case of higher value of (β). The conclusion drawn from Figs. 12, 13 if rate of infection is higher count of infectious sensor nodes are greater and susceptible sensor nodes are lower.

The impact of infection rate is presented with the help of Fig. 14. It is noticed from Fig. 14 that as the rate of infection increases, the count of infectious sensor node increases in the network. The count of infectious sensor nodes reaches to the maximum then begin to decline and attain the steady state or become zero; this depends on the value of basic reproduction number R_0^{th}. The value of R_0^{th} depends on other parametric values including the rate of infection. In case of higher value of (β), transmission rate of malware is also high in comparison to lower value.

The impact of (β) on exposed sensor nodes and infectious sensor nodes is plotted in Fig. 15. If value of (β) is higher, then count of exposed as well as infectious sensor nodes is higher. Because exposed nodes converted into infectious nodes after a certain interval of time that time is known as latent period. The latent period is used to control the transmission of malware in the network by applying the corrective action on time. When the node is in exposed state, it

Fig. 19 Impact of charging on recovery of sensor nodes

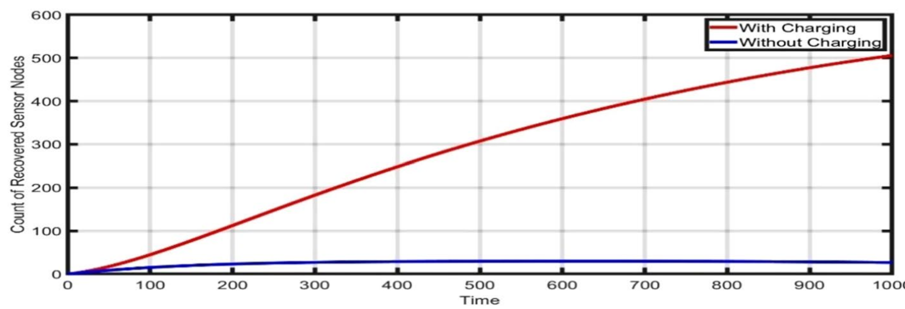


Fig. 20 Impact of charging on infectious and recovered nodes

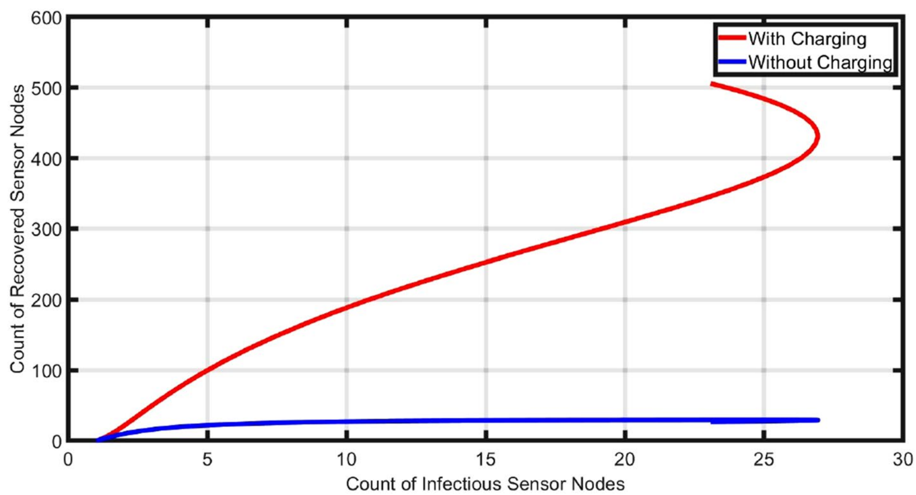
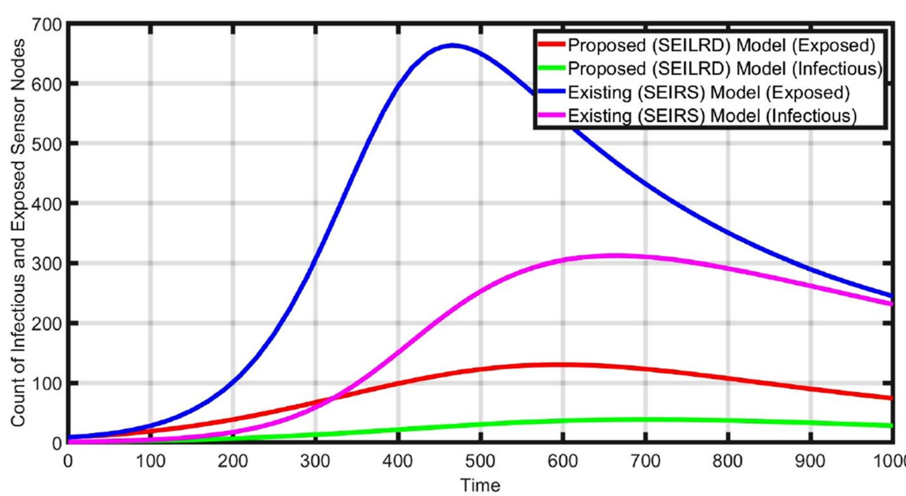


Fig. 21 Communication radial distance r=1.1



can be detected early in exposed state and apply preventive action on them to stop transmission.

The impact of recovery rate on exposed state of sensor nodes is explained with the help of Fig. 16. If the value τ is fixed and changes the value of ω from 0.003 to 0.007, as the value of ω increases count of infectious nodes increases and if value of ω is fixed and value of τ increases from 0.003 to 0.005. It is found that as the value of τ increases, count of recovered also increases. This indicates that if the rate of conversion from exposed state to infectious state is high, then a greater number of nodes converted into infectious state. The corrective method applies in exposed state of nodes to prevent the malware transmission in WSN. If the execution rate of anti-malware is high, the lesser count of exposed nodes is converted into infectious state.

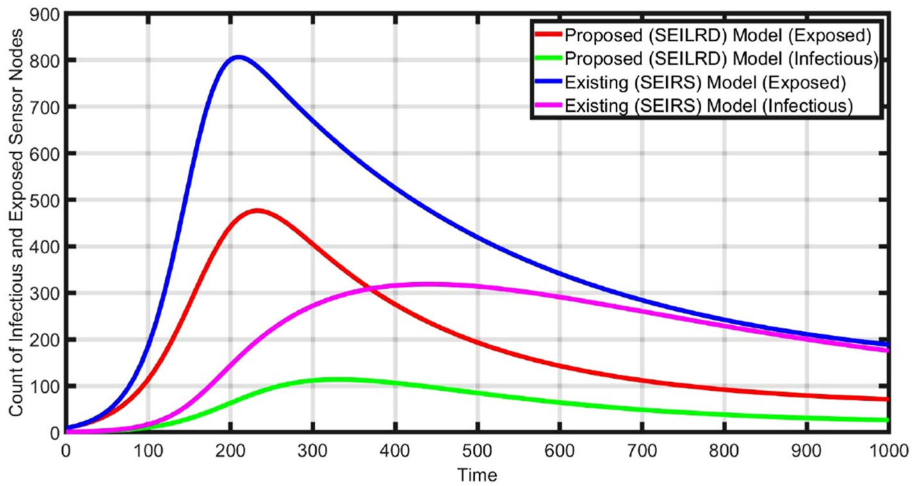


Fig. 22 Communication radial distance r = 2.2

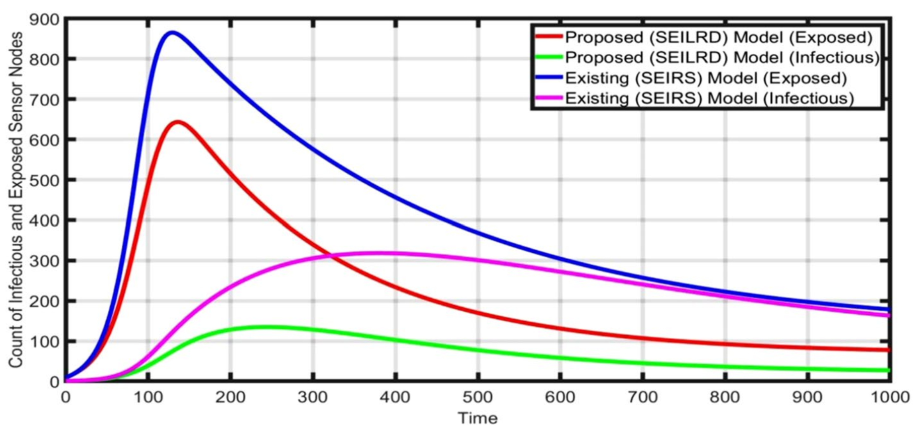


Fig. 23 Communication radial distance r = 3.4

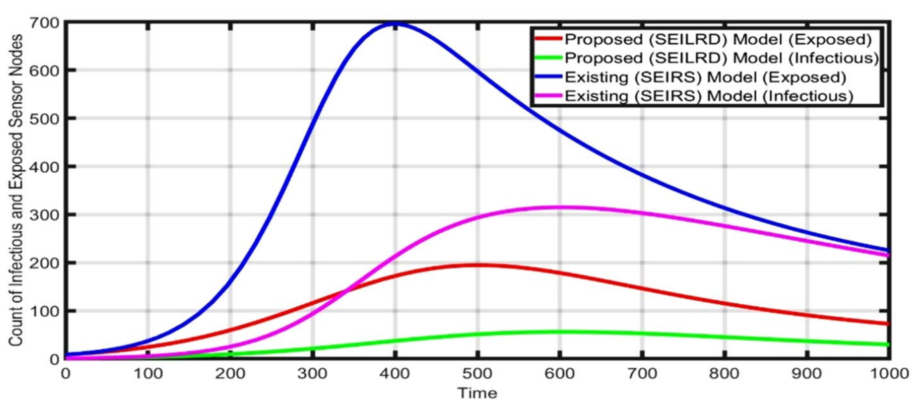


Fig. 24 Node density d = 8

The impact of various parameters is discussed in Figs. 17, 18 such as infection rate, charging rate, i.e., from low-energy state to recovered state, from infectious state to low-energy state, etc. From Fig. 17, it is observed that if the rate of infection is high, then initially count of infectious sensor nodes increases attain the maximum count and then starts to decrease due effect of charging and rate of recovery.

Figure 17 supports the result of Fig. 17. The larger number of infectious sensor nodes converted into recovered state when rate of recovery and charging rate of sensor nodes are high. The figure also illustrates the importance of charging sensor nodes with low-energy states. The effect of charging of low-energy sensor node is illustrated in Figs. 19, 20. Consider the case of charging and without charging of low-energy sensor nodes. It is found from Fig. 19 that when applying the mechanism of charging of sensor nodes, count of recovery nodes increases with time in respect to without charging. And Fig. 19 shows that lesser number of nodes reside in infectious state in case of charging. The method charging enhances the network lifetime and overcomes the problem of operational overhead.

Now, we perform a comparative analysis between the proposed model and the SEIRS model, existing model. For comparative analysis, keep the all parameters same in both the cases and vary the communication radius r. For better connectivity design communication radius is a crucial design parameter. Figures 21, 22, 23 illustrate that the performance of proposed model is better with respect to the SEIRS existing model for the distinct value of r. The graph is plotted between the count of exposed and infectious nodes and time. The count of exposed sensor nodes is increasing as the size of the communication radius increases. A similar type of behavior is also shown in the case of infectious sensor nodes and exposed sensor nodes. In both the proposed and existing models, the dynamics of malware transmission in the network are similar, but in the proposed model, the increment of exposed and infectious nodes is smaller compared to the existing model. The proposed model suggests a technique for the design of sensor nodes through which a secure and strong connecting system can be developed. Therefore, the proposed model suggests a better mechanism for controlling malware transmission in WSN.

Fig. 25 Node density d = 11

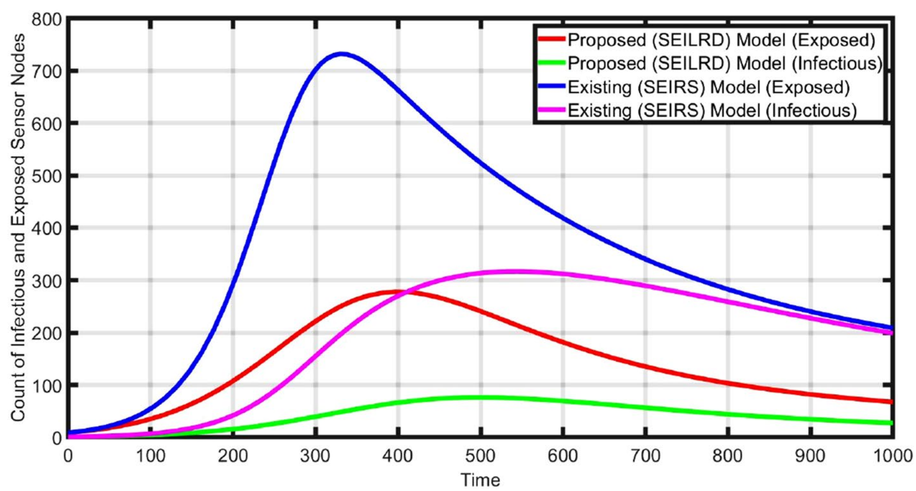
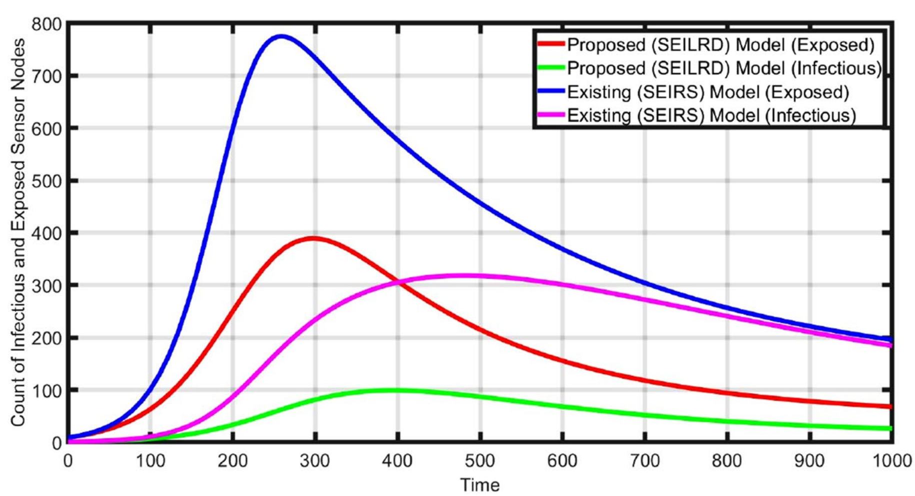


Fig. 26 Node density d = 17



Figures 24, 24, 26 illustrate that the performance of proposed model is better with respect to the SEIRS [15] existing model for the distinct value of node density d. The graph is plotted between the count of exposed and infectious nodes and time. The count of exposed sensor nodes is increasing as the value of node density is increasing. A similar type of behavior is also shown in the case of infectious sensor nodes and exposed sensor nodes. In both the proposed and existing models, the dynamics of malware transmission in the network are similar, but in the proposed model, the increment of exposed and infectious nodes is smaller compared to the existing model. The model provides the method of deployment of sensor nodes in the sensor field in an optimal manner. Therefore, the proposed model suggests a better mechanism for controlling malware transmission in WSN.

Figures 21, 22, 23, 24, 25, 26 show that counts of infectious nodes in both the models are not approaching to zero, this is the state of endemic equilibrium. Additionally, due to non-zero rate of infection, the peaks of both exposed and infectious sensor nodes are substantially greater. When an attack occurs in the system, then an appropriate and pertinent security mechanism is needed to exterminate malware from the network for that purpose the mechanism applies through the proposed model. The model used the concept of exposed state and charging of low-energy state nodes.

Conclusion

This work has proposed an enhanced epidemic SEILRD model with the primary goal of controlling the spread of malware and enhancing the lifetime of WSN. It is an extension of SILRD model. In addition, the exposed state in the model provides a mechanism to detect the movement of sensor nodes from a vulnerable status to an infected status during the early stages of the infection process. Such detection provides an opportunity to further implement methods to maintain the sensor nodes and stabilize WSN under different conditions. An improved SEILRD model, which utilizes charging as its primary method of operation, is able to eliminate the drawbacks of the traditional SILRD model. The maintenance of the system, such as the detection and removal of malicious software, is performed, while the sensor node is in sleep mode. The anti-malware potential of the network has been improved as a result of the proposed model; the network is now capable of perfectly and reliably adapting to new and different kinds of malware. The proposed model does not call for any additional hardware or signaling overhead to be implemented R_0^{th} is one of the critical criteria, which play an important role in controlling of malware transmission in WSN. Since overage and connectivity are also essential criteria in WSN, it is necessary to have enhanced SEILRDs. Moreover, discussed that malware-free equilibrium is locally and globally asymptotically stable if the value of $R_0^{th} \leq 1$ network unstable condition is $R_0^{th} > 1$. The endemic stability (if the value of $R_0^{th} > 1$) is also explained by the simulation results. For better design of WSN computed the threshold value of communication radius, which help in controlling of malware transmission and lifetime enhancement of WSN. Calculating the threshold value of node density is necessary for this purpose, because it provides information on the number of nodes that need to be deployed in the sensor fields to ensure smooth and secure communication while also optimizing the cost of deployment. This is yet another important concern regarding the deployment of sensor nodes in the sensor field. Article also discussed the relationship of R_0^{th} with r and d. In addition, the effects of charging and the significance of low-energy states are broken down in detail. Additionally, for the validity of the proposed model, the simulation was carried out using MATLAB (R2018a). The effect of different parameters on the transmission of malware in WSN under different conditions has been meticulously analyzed. The comparative evaluation between SEIRS and SEILRD has been performed. The comparative analysis shows a lower count of exposed and infectious sensor nodes in the proposed SEILRD model compared to the existing SEIRS model. Extensive outcomes provide evidence that the proposed model ensures increased network lifetime and better security mechanisms against malware attacks.

Declarations

Conflict of interest The manuscript has been approved by all authors and has never been published, or under consideration for publication elsewhere. We have not submitted our manuscript to a preprint server prior to submission on SN Computer Science.

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Authors and Affiliations

Shashank Awasthi[1,2] · Pramod Kumar Srivastava[3] · Naresh Kumar[2] · Rudra Pratap Ojha[2] · Ashok Kumar Yadav[4]

✉ Pramod Kumar Srivastava
drpks@gecazamgarh.ac.in

Shashank Awasthi
shashankglbitm@gmail.com

Naresh Kumar
naresh.dhull@gmail.com

Rudra Pratap Ojha
rpojha@gmail.com

Ashok Kumar Yadav
ashokyadav88.jnu@gmail.com

[1] Research Management Center, Management and Science University, 40100 Shah Alam, Selangor Darul Ehsan, Malaysia

[2] Department of Computer Science & Engineering, G.L.Bajaj Institute of Technology & Management, Street, Greater Noida, Uttar Pradesh 100190, India

[3] Department of Mathematics, Rajkiya Engineering College Azamagrh, Deogaon, Azamagrh, Uttar Pradesh 276201, India

[4] Department Information Technology, Rajkiya Engineering College Azamagrh, Deogaon, Azamagrh, Uttar Pradesh 276201, India